Statement of Work

StreamFlix Streaming Infrastructure Modernization

Field	Detail
Client	StreamFlix
Engagement	Cloud migration and platform modernization
Prepared by	Solutions / Delivery team
Status	Draft for review
Document basis	Discovery transcript and scope assessment

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1. Executive Summary

StreamFlix is experiencing rapid subscriber growth, particularly across international markets, that its current on-premises infrastructure can no longer sustain. Peak demand events produce buffering and outages, including a recent 30-minute failure during a season finale, and the operations team is consumed by manual firefighting rather than feature delivery.

This Statement of Work defines the engagement to migrate StreamFlix to a multi-region, cloud-native, microservices platform that scales automatically with demand. The solution delivers low-latency video playback, a foundation for live sports streaming, an improved recommendation and search experience, and stronger resilience, security, and observability.

The engagement targets readiness within 8 months, ahead of StreamFlix's planned live sports launch, at an initial migration investment of $2 to $3 million and an expected steady-state operating cost of approximately $400,000 per month. It begins with a low-risk assessment and proof-of-concept phase before progressively migrating production services.

2. In-Scope and Out-of-Scope

2.1 In-Scope

- Multi-region cloud landing zone and microservices platform on Kubernetes, with serverless functions for thumbnail generation and metadata processing
- Global content delivery via cloud CDN with edge locations to minimize latency
- Live streaming foundation: WebRTC delivery with RTMP ingestion, cloud-native transcoding, and adaptive bitrate streaming
- Real-time recommendation engine using stream processing and machine learning
- Improved search: full-text with fuzzy matching, personalized ranking, auto-complete, faceted filtering, and multi-language support
- Content moderation service for user-generated reviews and comments
- Data platform: read replicas, caching, time-series store, data warehouse, and message queues supporting real-time, batch, and reporting access
- Automation of content ingestion/transcoding pipelines and the release process
- CI/CD modernization, automated testing (unit, integration, end-to-end with device farms), and blue-green deployments
- Observability stack with real-time alerting and user-experience monitoring
- Security controls: encryption at rest and in transit, DRM, MFA for admin access
- Disaster recovery: multi-region active-active with real-time replication and automated traffic rerouting
- Internationalization: multi-language CMS, geofencing, and regional content restrictions
- Knowledge transfer and team enablement on cloud technologies

2.2 Out-of-Scope

- Virtual watch parties feature (raised but not yet specified; candidate for a future phase)
- Final selection of a specific cloud provider, pending assessment
- Recruitment of StreamFlix's planned SRE hires (StreamFlix responsibility)
- Production of original content, licensing negotiations, and content acquisition
- Client application redesign beyond integration with new streaming and search services
- Chaos engineering at full regional-outage scale during initial phases (introduced incrementally)
- Work beyond the first three months is indicative only and confirmed in later planning

3. Business Objectives

Stated business priorities, in order: customer satisfaction (top priority), operational cost reduction, and release velocity. The objectives below are tracked against agreed KPIs.

Objective	Target / Measure
Eliminate peak-load instability	Maintain streaming quality during peak hours; remove manual peak intervention
Improve playback experience	Sub-2-second video startup time; reduced buffering ratio and playback failures
Enable live sports streaming	Operational live streaming with under 10 seconds latency from the live feed
Increase resilience	RPO reduced from 24 hours to 1 hour; RTO under 15 minutes for critical services
Reduce operational toil	Higher deployment frequency; lower mean time to resolution; automated content and release workflows
Control and attribute cost	Operating cost near $400,000/month; cost reporting by department and feature
Grow and retain subscribers	Improved subscriber retention, engagement rates, and revenue per user

4. Technical Solution

4.1 Architecture

A multi-region cloud deployment built on a microservices architecture. Most services run as containers orchestrated by Kubernetes; serverless functions handle discrete tasks such as thumbnail generation and metadata processing. An API Gateway fronts client traffic, with GraphQL serving differing client requirements.

4.2 Content Delivery and Streaming

- Cloud provider CDN with global edge locations for low-latency delivery
- Live streaming via WebRTC with RTMP ingestion for sports content
- Cloud-native transcoding and adaptive bitrate streaming across mobile, web, and smart TV

4.3 Recommendations and Search

- Real-time recommendations from a streaming pipeline feeding analytics and ML services
- Event sourcing and CQRS for user behavior and recommendation read/write separation
- Search via Elasticsearch: fuzzy full-text, personalized ranking, auto-complete, faceted and multi-language search

4.4 Data Platform

- Read replicas for analytics; caching layers for hot content
- Time-series database for metrics and behavior; data warehouse for business intelligence
- Message queues for asynchronous processing
- Hybrid analytics: real-time streaming for live metrics, batch for historical, materialized views and event-driven dashboards

- Retention: user data indefinite; viewing history archived after 2 years

4.5 Resilience Patterns

- Circuit breaker for external integrations
- Saga pattern for distributed transactions, especially payments
- Multi-region active-active with real-time data replication and automated failover routing

4.6 Technology Stack

Aligned to team expertise: Node.js for API services and Python for machine learning, introducing additional technologies where justified.

- API layer: GraphQL
- Caching and search: Redis and Elasticsearch
- Event streaming: Apache Kafka
- Observability: Prometheus and OpenTelemetry

4.7 Automation and Delivery

- Automated content ingestion, parallel transcoding with quality checks, ML metadata extraction, compliance validation
- Release automation: testing gates, progressive rollouts, automated rollback, performance validation
- Cost optimization: demand-based auto-scaling, spot instances for encoding, storage tiering, reserved baseline capacity, cost allocation tags

5. Security and Compliance

Area	Requirement
Encryption	At rest and in transit across all data stores and transport
Content protection	DRM support for licensed video content
Access control	Multi-factor authentication mandatory for the admin portal
GDPR	Compliance required for European market expansion
PCI	Compliance maintained for payment processing
Regional restrictions	Geofencing and regional content controls per licensing agreements
Moderation	Multi-layer review of user-generated content: ML first pass, human queue, rate limiting, reputation, appeals

6. Timeline

Target readiness within 8 months, ahead of the live sports launch. The first three months are defined below; subsequent phases cover broader production migration, live streaming build-out, and hardening, confirmed in later planning.

Phase	Activities
Month 1	Detailed architecture design; core infrastructure and CI/CD setup; knowledge transfer and training; proof of concept on a non-critical service (e.g. metadata)
Month 2	Build foundational services; implement monitoring and alerting; establish security controls; create development and staging environments
Month 3	Migrate first production services; implement automated testing; fine-tune performance; document operational procedures

Months 4 to 8	Progressive production migration; live streaming and recommendations build-out; resilience testing; cost and performance optimization toward launch readiness

7. Risks

Risk	Impact	Mitigation
Fixed launch deadline	High	Phased delivery starting with low-risk PoC; prioritize live streaming critical path early
Limited team cloud experience	Medium	Knowledge transfer, training, and PoC on a non-critical service before production migration
SRE roles not yet filled	Medium	Sequence migration around planned hires; embed delivery support during ramp-up
Cloud cost overruns	Medium	Auto-scaling, spot/reserved instances, tiering, and cost allocation tags with departmental reporting
Production migration of live services	High	Blue-green deployments, automated rollback, progressive rollouts, chaos engineering introduced incrementally
Compliance gaps (GDPR, PCI, DRM)	High	Security and compliance controls built in from the start, not retrofitted
Data growth (~20TB/month)	Medium	Intelligent storage tiering and retention/archival policy for viewing history
Unscoped features (e.g. watch parties)	Low	Held out of current scope; assessed for a future phase to protect the core timeline